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# CS & DA

## ENGINEERING MATHEMATICS

DPP: 2

## CALCULUS AND OPTIMIZATION

**Q1** A 2m ladder leans against a vertical wall. If the top of the ladder begins to slide down the wall at the rate 25 cm/sec. then the rate (in cm/sec.) at which the bottom of the ladder slides away from the wall on the horizontal ground when the top of the ladder is 1 m above the ground is:

(A) $25\sqrt{3}$ (B) 25

(C) $\frac{25}{\sqrt{3}}$ (D) $\frac{25}{3}$

**Q2** The percentage error in calculating the volume of a cubical box if an error of 1%is made in measuring the length of edges of the cube is :

(A) 1% (B) 2%

(C) 3% (D) 6%

**Q3** Find the tangent line to f (x) = 7x + $4e^x$ at x = 0.

**Q4** Differentiate $f(t) = \frac{1+5t}{\ln(t)}$

**Q5** Find $\frac{\partial w}{\partial x}$ for the following function.

$$w = \cos\left(x^2 + 2y\right) - e^{4x - z^4 y} + y^3$$

**Q6** Find all the 1st order partial derivatives of the following function.

$$f(x, y, z) = 4x^3y^2 - e^z y^4 + \frac{z^3}{x^2} + 4y - x^{16}$$

**Q7** Prove that

f (x, y) = $x^3 - 2x^2y + 3xy^2 + y^3$

is homogeneous; what is the degree? Verify Euler's Theorem for f .

**Q8** If $v(x, y) = \log\left(\frac{x^2+y^2}{x+y}\right)$, prove that $x\frac{\partial v}{\partial x} + y\frac{\partial v}{\partial y} = 1$

**Q9** **Prove that**

**g(x, y) = x log (y/x)**

**is homogenous, what is the degree ? Verify Euler's theorem for g.**

**Q10** The jacobian of p, q, r, w. r. t x, y, z given = p = x + y + z, q = y + z, r = z is _____.

(A) 0 (B) 1

(C) 2 (D) $-1$

**Q11** Given $u = \frac{yz}{x}, v = \frac{zx}{y}, w = \frac{xy}{z}$ then the value of $\frac{\partial(u,v,w)}{\partial(x,y,z)}$ is _____.

(A) 4 (B) − 4

(C) 0 (D) 1

**Q12** The function f(x) = 2log (x−2) − $x^2$ + 4x + 1 increase on the interval

(A) (1, 2) (B) (2,3)

(C) (5/2, 3) (D) (2, 4)

**Q13** The function $f(x) = \frac{x^2-1}{x^2+1}$ :

(A) is concave for $x \in \left[\frac{-1}{\sqrt{3}}, \frac{1}{\sqrt{3}}\right]$

(B) is convex for $x \in \left[\frac{-1}{\sqrt{3}}, \frac{1}{\sqrt{3}}\right]$

(C) has 2 points of inflexion

(D) is concave for $\left[\frac{1}{\sqrt{3}}, \infty\right)$

**Q14** Number of point of inflections of the function:

$f(x) = x^4 - 18x^2 + 9$

(A) 1 (B) 2

(C) 3 (D) 4

**Q15** The function f(x) = $2x^3$ − $3x^2$ − 36x + 2 has its maxima at

(A) x = − 2 only

(B) x = 0 only

(C) x = 3 only

(D) both x = −2 and x = 3

**Q16** The maximum value of f(x) = $x^3$ −$9x^2$ + 24x + 5 in the interval [1, 6] is

**Q17** The minimum value of the function $f(x) = x^3 - 3x^2 - 24x + 100$ in the interval

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[-3, 3] is

**Q18** Suppose that the amount of money in a bank account after t year is

given by $A\left(t\right) = 2000 - 10te^{5-\frac{t^2}{8}}$

The minimum and maximum amount of money in the account during the first 10 year that it is open occur respectively at:

(A) T = 2, t = 0 (B) T = 2, t = 10

(C) T = 0, t = 2 (D) T = 10, t = 0

**Q19** The function: $f(x) = x^2 \ln(3x) + 6$ has:

(A) No critical point

(B) Only one critical point

(C) Only one stationary point

(D) Infinitely many critical points

**Q20** The function $f(x,y) = x^2y - 3xy + 2y + x$ has

(A) No local extremum

(B) One local minimum but no local maximum

(C) One local maximum but no local minimum

(D) One local minimum and one local maximum

**Q21** Find the local minima of the function f(x , y) = $2x^2 + 2xy + 2y^2 - 6x$

(A) (2,1) (B) (-2,1)

(C) (2,0) (D) (2,-1)

**Q22** Find the critical points of the function f(x, y) = $y^2 - x^2$ and check for the presence of any saddle point.

**Q23** Find the critical points of the function f(x, y) = $x^2y^2 - 5x^2 - 5y^2 - 8xy$ and check for the presence of any saddle point.

**Q24** Calculate the perpendicular distance between origin and a point P on surface of curve $z^2 = 3xy + 4$

**Q25** A jet of an enemy is flying along the curve $y = x^2 + 2$. A soldier is placed at the point (3,2). What is the nearest distance between the soldier and the jet?

**Q26** A Square piece of tin of side 24 cm is to be made into a box without top by cutting a square from each corner and folding up the flaps to from a box. What should be the side of the square to be cut off so that the volume of the box is maximum ? Also find this maximum volume.

**Q27** Manufacturer can sell x items at a price of rupees $\left(5 - \frac{x}{100}\right)$ each. The cost price is Rs. $\left(\frac{x}{5} + 500\right)$. Find the number of items he should sell to earn maximum profit.

**Q28** Evaluate $\int x^2 \sin x \, dx$

**Q29** $\int \frac{dx}{2\sqrt{x}(x+1)}$

**Q30** Evaluate: $\int \frac{dx}{(x-1)(x-2)(x-3)}$

**Q31** Evaluate: $\int_2^7 \frac{\sqrt{x}dx}{\sqrt{x}+\sqrt{9-x}}$

**Q32** Evaluate: $\int_{-\sqrt{3}}^{\sqrt{3}} \frac{dx}{(1+e^x)(1+x^2)}$

**Q33** Evaluate: $\int_{-a}^{a} \sqrt{\frac{a-x}{a+x}}dx$

**Q34** Evaluate: $\int_0^{\pi} \frac{xdx}{1+\cos^2 x}$

**Q35** Evaluate: $\int_0^{\pi} \frac{dx}{1+2\sin^2 x}$

**Q36** Let $5f(x) + 4f\left(\frac{1}{x}\right) = \frac{1}{x} + 3, x > 0$ Then $18\int_1^2 f(x)dx$ is equal to:

(A) 10 log 2 – 6

(B) 10 log 2 + 6

(C) 5 log 2 – 6

(D) 5 log 2 – 3

**Q37** $If\ \mathrm{F}(x) = \int_x^{x^2} \sqrt{\sin t}dt,$ then find $\mathrm{F}'(x)$

**Q38**

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Evaluate: $\lim_{x\to\infty} \frac{\left(\int_0^x e^{t^2}\,dt\right)^2}{\int_0^x e^{2t^2}\,dt}$

**Q39** Arc length of the curve $y = x^{3/2}$ , $z = 0$ from (0,0,0) to (4, 8, 0) is-

(A) $\frac{8}{27}\left(10^{3/2}+1\right)$

(B) $\frac{8}{27}\left(10^{3/2}-2\right)$

(C) $\frac{8}{27}\left(10^{3/2}-1\right)$

(D) $\frac{8}{27}\left(10^{3/2}+2\right)$

**Q40** Let f be an increasing, differentiable function. If the curve y =f(x) passes through (1, 1) and has length $L=\int_1^2 \sqrt{\left(1+\frac{1}{4x^2}\right)}dx, 1\le x\le 2,$ then the curve is-

(A) $y=\ln\left(\sqrt{x}\right)-1$ (B) $y=1-\ln\left(\sqrt{x}\right)$

(C) $y=\ln\left(1+\sqrt{x}\right)$ (D) $y=1+\ln\left(\sqrt{x}\right)$

**Q41** The volume of the solid of revolution of $y=\frac{a}{2}\left(e^{x/a}+e^{-x/a}\right)$ about x-axis between x= 0 and x= b is-

(A) $\frac{\pi a^3}{8}\left(e^{2b/a}-e^{-2b/a}\right)-\frac{\pi a^2 b}{2}$

(B) $-\frac{\pi a^3}{8}\left(e^{2b/a}-e^{-2b/a}\right)+\frac{\pi a^2 b}{2}$

(C) $-\frac{\pi a^3}{8}\left(e^{2b/a}-e^{-2b/a}\right)-\frac{\pi a^2 b}{2}$

(D) $\frac{\pi a^3}{8}\left(e^{2b/a}-e^{-2b/a}\right)+\frac{\pi a^2 b}{2}$

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# Answer Key

**Q1** (C)

**Q2** (C)

**Q3** 5.9459.

**Q4** $\boxed{\frac{5\,\mathrm{In(t)}-\frac{1}{t}-5}{[\mathrm{In(t)}]^2}}$

**Q5** $-2x\sin\left(x^2+2y\right)-4e^{4x-z^4y}$

**Q6** $\frac{\partial f}{\partial y}=f_z=-e^z y^4+\frac{3z^2}{x^2}$

**Q7** 3

**Q8** 1

**Q9** 1

**Q10** (B)

**Q11** (A)

**Q12** (B, C)

**Q13** (A, C)

**Q14** (B)

**Q15** (A)

**Q16** 25

**Q17** 28

**Q18** (A)

**Q19** (B, C)

**Q20** (A)

**Q21** (D)

**Q22** (0,0)

**Q23** Hence, at (±1, ±1) also we have saddle points.

**Q24** 2

**Q25** $\sqrt{5}$

**Q26** 1024cm$^3$

**Q27** 240

**Q28** $-x^2\cos x+2x\sin x+2\cos x+c$

**Q29** $\tan^{-1}(\sqrt{x})+c$

**Q30** $\ln\left(\frac{\sqrt{x^2-4x+3}}{|x-2|}\right)+c$

**Q31** $I=\frac{5}{2}$

**Q32** $\frac{\pi}{3}$

**Q33** $a\left[\frac{\pi}{2}-0\right]=\pi a$

**Q34** $\frac{\pi^2}{2\sqrt{2}}$

**Q35** $2\frac{1}{\sqrt{3}}\left[\tan^{-1}t\sqrt{3}\right]_0^{\infty}=\frac{2}{\sqrt{3}}\times\frac{\pi}{2}=\frac{\pi}{\sqrt{3}}.$

**Q36** (A)

**Q37** $F'(x)=2x\cdot\sqrt{\sin x^2}-1\cdot\sqrt{\sin x}$

**Q38** 0

**Q39** (C)

**Q40** (D)

**Q41** (D)

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# Hints & Solutions

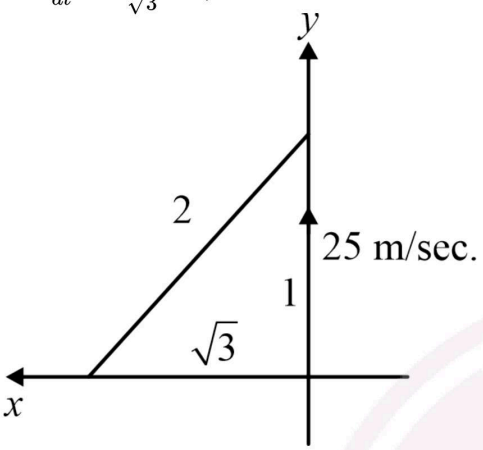
**Q1 Text Solution:**

$x^2 + y^2 = 4$

$x \times \frac{dx}{dt} + y\frac{dy}{dt} = 0$

$\Rightarrow \sqrt{3}\frac{dx}{dt} - 1\,(25) = 0 \left(\frac{dy}{dt} = -25\right)$

$\Rightarrow \frac{dx}{dt} = \frac{25}{\sqrt{3}} cm/\sec.$




**Q2 Text Solution:**

Let x be the length of an edge of the cube and y be its volume. Then, $y = x^3$. Let $\Delta x$ be the error in x and y be the corresponding error in $\Delta y$.

Then,

$\frac{\Delta x}{x} \times 100 = 1$ (given)

$\Rightarrow \frac{dx}{x} \times 100 = 1$ [as $dx \cong \Delta x$] ...(i)

We have to find $\frac{\Delta y}{y} \times 100$

Now, $y = x^3 \Rightarrow \frac{dy}{dx} = 3x^2$

$\therefore \quad dy = \frac{dy}{dx}dx \Rightarrow$ **$dy = 3x^2 . dx$**

$\Rightarrow \frac{dy}{y} = \frac{3x^2}{y}dx$

$\Rightarrow \frac{dy}{y} = \frac{3x^2}{x^3}dx$ [as $y = x^3$]

$\Rightarrow \frac{dy}{y} = 3\frac{dx}{x}$

$\Rightarrow \frac{dy}{y} \times 100 = 3\left(\frac{dx}{x} \times 100\right) = 3$ [using (i)]

$\Rightarrow \frac{\Delta y}{y} \times 100 = 3$ [as $dy = \Delta y$]

So there is 3% error in calculating the volume of the cube.

**Q3 Text Solution:**

We know that the derivative of the function will give us the slope of the tangent line so we'll need the derivative of the function.

f' (x) = 7x ln (7) + 4ex

Now all we need to do is evaluate the function and the derivative at the point in question.

f (0) = 5, f' (0) = ln (7) + 4 = 5.9459.

**Q4 Text Solution:**

$f\left(t\right) = \frac{g(t)}{h(t)}$

$differentiating\ by\ u \sin g\ the\ u/v\ formula$

$f'\left(t\right) = \frac{h(t).g'(t) - g(t).h'(t)}{h^2(t)}$

$f'\,(t) = \frac{5\,\ln(t) - (1+5t)\left(\frac{1}{t}\right)}{[\ln(t)]^2} = \boxed{\frac{5\,\ln(t) - \frac{1}{t} - 5}{[\ln(t)]^2}}$

**Q5 Text Solution:**

$\frac{\partial w}{\partial x} = w_x = -2x\sin\left(x^2 + 2y\right) - 4$

$e^{4x - z^4 y}$

**Q6 Text Solution:**

$\frac{\partial f}{\partial x} = f_x = 12x^2y^2 - \frac{2z^3}{x^3} - 16x^{15}$

$\frac{\partial f}{\partial y} = f_y = 8x^3y - 4e^z y^3 + 4$

$\frac{\partial f}{\partial y} = f_z = -e^z y^4 + \frac{3z^2}{x^2}$

**Q7 Text Solution:**

$f(x, y) = x^3 - 2x^2y + 3xy^2 + y^3$

Apply $x = \lambda x$ and $y = \lambda y$

$f(\lambda x, \lambda y) = (\lambda x)^3 - 2(\lambda x)^2(\lambda y) + 3(\lambda x)(\lambda y)^2 + (\lambda y)^3$

$f(\lambda x, \lambda y) = \lambda^3 x^3 - 2\lambda^3(x^2y) + 3\lambda^3 xy^2 + \lambda^3 y^3$

$f(\lambda x, \lambda y) = \lambda^3(x^3 - 2x^2y + 3xy^2 + y^3)$

So it is a homogeneous function of degree 3.

Given : $x^3 - 2x^2y + 3xy^2 + y^3$

**$\partial F/\partial x = 3x^2 - 2(2x)y + 3(1)y^2 + 0$**

**$\partial F/\partial x = 3x^2 - 4xy + 3y^2$**

**$x(\partial F/\partial x) = x(3x^2 - 4xy + 3y^2)$ ---(1)**

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Given : $x^3 - 2x^2y + 3xy^2 + y^3$

$\partial F/\partial y = 0 - 2x^2(1) + 3x(2y) + 3y^2$

$\partial F/\partial y = -2x^2 + 6xy + 3y^2$

$y(\partial F/\partial y) = y(-2x^2 + 6xy + 3y^2)$ ---(2)

(1) + (2)

$x(\partial F/\partial x) + y(\partial F/\partial y) = x(3x^2 - 4xy + 3y^2) + y(-2x^2 + 6xy + 3y^2)$

$= 3x^3 - 4x^2y + 3xy^2 - 2x^2y + 6xy^2 + 3y^3$

$= 3x^3 - 6x^2y + 9xy^2 + 3y^3$

$= 3(x^3 - 2x^2y + 3xy^2 + y^3)$

= P F(x, y)

Hence it is proved.

**Q8 Text Solution:**

Given that,

$v(x, y) = \log\left(\frac{x^2+y^2}{x+y}\right)$

Change into an exponential function.

Let $e^v = \frac{x^2+y^2}{x+y} = f(x, y)$

$f(x, y) = \frac{\lambda^2x^2+\lambda^2y^2}{\lambda x+\lambda y}$

$\Rightarrow f(x, y) = \frac{\lambda^2(x^2+y^2)}{\lambda(x+y)} f$ is a homogeneous fuction of degree 1.

By Euler' Theorem,

$x\frac{\partial f}{\partial x} + y\frac{\partial f}{\partial y} = 1 \times f$

Putting $f=e^v$

$\Rightarrow x\frac{\partial e^v}{\partial x} + y\frac{\partial e^v}{\partial y} = e^v$ exists.

Using chain rule-

$\Rightarrow xe^v \frac{\partial v}{\partial x} + ye^v \frac{\partial v}{\partial y} = e^v$

Hence Proved.

$\Rightarrow x\frac{\partial v}{\partial x} + y\frac{\partial v}{\partial y} = \frac{e^v}{e^v} = 1.$

**Q9 Text Solution:**

**g(x, y) = x log (y/x)**

Applying x = λx and y = λy

**g(λx, λy) = λx log (λy/λx)**

**g(λx, λy) = λx log (y/x)**

**It is a homogenous function of degree 1.**

**$\partial F/\partial x = x(x/y)(-yx^{-2}) + \log(y/x)(1)$**

**$\partial F/\partial x = x(x/y)(-y/x^2) + \log(y/x)$**

**$\partial F/\partial x = -1 + \log(y/x)$ -----(1)**

**$\partial F/\partial y = x(x/y)(1/x)$**

**$\partial F/\partial y = (x/y)$ -----(2)**

**(1) + (2)**

**$x(\partial F/\partial x) + y(\partial F/\partial y) = x[-1 + \log(y/x)] + y(x/y)$**

**$x(\partial F/\partial x) + y(\partial F/\partial y) = -x + x\log(y/x) + x$**

**$x(\partial F/\partial x) + y(\partial F/\partial y) = x \log(y/x)$**

**Hence it is proved.**

**Q10 Text Solution:**

We have to find

$$J = \frac{\partial(p,q,r)}{\partial(x,y,z)} = \begin{vmatrix} \frac{\partial p}{\partial x} & \frac{\partial p}{\partial y} & \frac{\partial p}{\partial z} \\ \frac{\partial q}{\partial x} & \frac{\partial q}{\partial y} & \frac{\partial q}{\partial z} \\ \frac{\partial r}{\partial x} & \frac{\partial r}{\partial y} & \frac{\partial r}{\partial z} \end{vmatrix}$$

But p = x + y + z, q = y + z, r = z (taking partial derivative)

$$J = \begin{vmatrix} 1 & 1 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{vmatrix}$$

$\left(\frac{\partial p}{\partial x} = 1, \frac{\partial p}{\partial y} = 1, \frac{\partial p}{\partial z} = 1, \frac{\partial q}{\partial x} = 0, \frac{\partial q}{\partial y} = 1, \frac{\partial q}{\partial z} = 1, \frac{\partial r}{\partial x} = 0, \frac{\partial r}{\partial y} = 0, \frac{\partial r}{\partial z} = 1,\right)$

$\frac{\partial r}{\partial y} = 0, \frac{\partial r}{\partial z} = 1$

On expanding we get

J = 1 (1 − 0 ) = 1

Thus J = 1.

**Q11 Text Solution:**

By Data

$u = \frac{yz}{x}, v = \frac{zx}{y}, w = \frac{xy}{z}$

$$J = \frac{\partial(u,v,w)}{\partial(x,y,z)} = \begin{vmatrix} \frac{\partial u}{\partial x} & \frac{\partial u}{\partial y} & \frac{\partial u}{\partial z} \\ \frac{\partial v}{\partial x} & \frac{\partial v}{\partial y} & \frac{\partial v}{\partial z} \\ \frac{\partial w}{\partial x} & \frac{\partial w}{\partial y} & \frac{\partial w}{\partial z} \end{vmatrix}$$

$$= \begin{vmatrix} \frac{-yz}{x^2} & \frac{z}{x} & \frac{y}{x} \\ \frac{z}{y} & \frac{-zx}{y^2} & \frac{x}{y} \\ \frac{y}{z} & \frac{x}{z} & \frac{-xy}{z^2} \end{vmatrix}$$

$$= \frac{-yz}{x^2}\left\{\left(\frac{-zx}{y^2}\right)\left(\frac{-xy}{y^2}\right) - \left(\frac{x}{z}\right)\left(\frac{x}{y}\right)\right\} - \left(\frac{z}{x}\right)\left\{\left(\frac{z}{y}\right)\left(\frac{-xy}{x^2}\right) - \left(\frac{y}{z}\right)\left(\frac{x}{y}\right)\right\} + \frac{y}{x} - \left\{\left(\frac{z}{y}\right)\left(\frac{x}{z}\right) - \left(\frac{y}{z}\right)\left(\frac{-zx}{y^2}\right)\right\}$$

$$= \frac{-yz}{x^2} - \left\{\frac{x^2}{yz} - \frac{x^2}{yz}\right\} - \frac{z}{x}\left\{\frac{-x}{z} - \frac{x}{y}\right\} + \frac{y}{x}\left\{\frac{x}{y} + \frac{x}{y}\right\} = 0 + 1 + 1 + 1 = 4$$

Therefore $\frac{\partial(u,v,w)}{\partial(x,y,z)} = 4.$

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**Q12 Text Solution:**

f (x) = 2 log (x −2)−$x^2$+4x+1

$\Rightarrow f'(x) = \frac{2}{x-2} - 2x + 4$

$f'(x) = \frac{2}{x-2} - 2x + 4$

$\frac{2-(2x-4)(x-2)}{x-2}$

$\frac{2-(2x^2-4x-4x+8)}{x-2}$

$\frac{-2x^2+8x-6}{x-2}$

$x - 2 = 0, x = 2$

$-2x^2 + 8x - 6 = 0$

$x^2 - 4x + 3 = 0$

$(x-1)(x-3) = 0$

$x = 1, 3$

Since,

$\therefore f'(x) > 0$

$\Rightarrow -2(x-1)(x-3)(x-2) > 0$

$\Rightarrow (x-1)(x-2)(x-3) < 0$

$\Rightarrow x \in (-\infty, 1) \cup (2, 3).$

$- \quad + \quad - \quad +$

$-\infty \quad 1 \quad 2 \quad 3 \quad \infty$

**Q13 Text Solution:**

Differentiate w.r.t x to get :

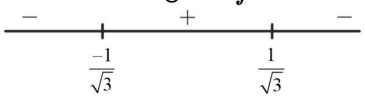
$f'(x) = \frac{2x(x^2+1)-2x(x^2-1)}{(x^2+1)^2} = \frac{4x}{(x^2+1)^2}$

Differentiate again to get :

$f''(x) = 4\frac{(x^2+1)^2 - x(4x)(x^2+1)}{(x^2+1)^4}$

$= 4\frac{(x^2+1)(1-3x^2)}{(x^2+1)^4} = \frac{-4(\sqrt{3x}-1)(\sqrt{3x}+1)}{(x^2+1)^3}$

Observe the sign of $f''$'(x) in the following figure




(using $x^2 + 1 > 0$)

Convex Convex Convex

From figure, points of inflextion are x = $\pm\frac{1}{\sqrt{3}}$

Intervals of concavity are x ∈ $\left[\frac{-1}{\sqrt{3}}, \frac{1}{\sqrt{3}}\right]$

Intervals of convexity are

$x \in \left(-\infty, \frac{-1}{\sqrt{3}}\right] \cup \left[\frac{1}{\sqrt{3}}, \infty\right).$

**Q14 Text Solution:**

$f(x) = x^4 - 18x^2 + 9$

$f'(x) = 4x^3 - 36x$

$f''(x) = 12x^2 - 36$

$f'''(x) = 24x$

*Points of inflection are −*

$f''(x) = 0, x = \sqrt{3}, -\sqrt{3}$

$f'''(\sqrt{3}), f'''(-\sqrt{3})$ *are both not equal to zero*

*Thus both are the points of inflection.*

**Q15 Text Solution:**

f(x) = $2x^3 - 3x^2 - 36x + 2$

f'(x) = 0

$6x^2 - 6x - 36 = 0$

$x^2 - x - 6 = 0$

(x − 3) (x + 2) = 0

x = 3 , −2

f'(x) = 12x − 6

at x = −2 f"(x) < 0

at x = 3 f"(x) > 0

Thus maximum at x=-2

**Q16 Text Solution:**

f(x) = $x^3 - 9x^2 + 24x + 5$

$f'(x) = 3x^2 - 18x + 24$

f(x) = 0

$3x^2 - 18x + 24 = 0$

$x^2 - 6x + 8 = 0$

$(x-4)(x-2) = 0$

x = 2 , 4

$f''(x) = 8x - 18$

$f''(x) = \text{at } x = 2$

$12 - 18 = 0$

thus maximum

$f''(x) \text{is} = -\text{ve at } x = 2$

at x = 6 thus

f(x) = $8 - 9 \times 4 + 24 \times 2 + 5$

$= 8 - 36 + 48 + 5$

= 13 + 48 −36

$= 13 + 12 = 25$

**Q17 Text Solution:**

$f(x) = x^3 - 3x^2 - 24x + 100$

$f'(x) = 3x^2 - 6x - 24$

$f'(x) > 0$

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$3x^2 - 6x - 24 = 0$

$x^2 - 2x - 8 = 0$

$(x-4)(x+2) = 0$

x = 4 , − 2

$f''(x) = 6x - 6$

$6 \times 4 - 6 > 24 - 6 < 0 \rightarrow max$

$f''(x) = 6x - 6$

$= 6x - 2 - 6 > -12 - 6 = 16 < 0$




$(-\infty, -2) \rightarrow -f\,x\,\text{is} + \text{ve}(\uparrow)$

$(4, \infty) \rightarrow f\,x\,\text{is} + \text{ve}(\uparrow)$

(- 2, 1) it is decreasing

Thus b/w (-2 ,4) the fx is decreasing

thus at x = 3 it will be min

$f(x) = 27 - 27 - 24 \times 3 + 100$

$= -72 + 100$

= 28

**Q18 Text Solution:**

We'll first need the derivative so we can find the critical points.

$A'(t) = -10e^{5-\frac{t^2}{8}} - 10te^{5-\frac{t^2}{8}}\left(-\frac{t}{4}\right)$

$= 10e^{5-\frac{t^2}{8}} - \left(-1 + \frac{t^2}{4}\right)$

Therefore the derivative will only by zero where,

$-1 + \frac{t^2}{4} = 0 \Rightarrow t^2 = 4$

$\Rightarrow t = \pm 2$

However only ***t*** = 2 is actually in the interval so that is only critical point that we'll use. Now for absolute maxima. We have, Max (***A*** (0), ***A*** (2), ***A*** (10)} i.e., Max (2000, 199.66, 1999.94} On comparing all these values we found, the maximum amount in the account will be Rs. 2000 which occurs at ***t*** = 0 and similarly absolute minimum amount in the account will be Rs.199.66 which occurs at the 2 year mark.

**Q19 Text Solution:**

$f'(x) = 2x\,\text{In}\,(3x) + x^2\left(\frac{3}{3x}\right)$

= 2x In (3x) + x

= x (2 In (3x) + 1)

The derivative will only be zero if,

2 In (3x) +1 = 0

In (3x) = $-\frac{1}{2}$

Take exponent on both sides to get :

$e^{\text{In}}(3x) = e^{-1/2}$

$3x = e^{-1/2}$

$x = \frac{1}{3}e^{-1/2} = \frac{1}{3\sqrt{e}}$

There is only one critical point of ***f*** (***x***)

i/e. $x = \frac{1}{3\sqrt{e}}$

Which is also a stationary point of ***f*** (***x***).

**Q20 Text Solution:**

$r=\partial^2 f/\partial x^2=2y$

$s=\partial^2 f/\partial x\partial y=2x-3$

$t=\partial^2 f/\partial y2=0$

Since, $rt-s^2 \leq 0$, (if $rt-s^2 < 0$ then we have no maxima or minima, if = 0 then we can't say anything).

Maxima will exist when $rt-s^2>0$ and $r<0$.

Minima will exist when $rt-s^2>0$ and $r>0$.

**As $rt-s^2$ is never greater than 0 so we have no local extremum.**

**Q21 Text Solution:**

$f_x(x,y) = 4x + 2y - 6=0 \quad (1)$

$f_y(x,y) = 2x + 4y=0 \quad (2)$

On solving (1) and (2) we get,

**x=2,y=-1**

$r=\partial^2 f/\partial x^2=4$

$s=\partial^2 f/\partial x\partial y=2$

$t=\partial^2 f/\partial y^2=4$

$rt-s^2=12$

**As $rt-s^2>0$ and $r>0$. Thus, (2,-1) is the point of local minima.**

**Q22 Text Solution:**

Let us find the first-order partial derivative with respect to x and y to determine the critical points.

$f_x(x, y) = -2x$

and $f_y(x, y) = 2y$ which exists everywhere.

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Clearly, f has a critical point at (0, 0). Let us apply the second-order partial derivative test,

$f_{xx}(x, y) = -2$

$f_{yy}(x, y) = 2$

$f_{xy}(x, y) = 0$

Therefore, the discriminant $D = f_{xx}(0, 0)\, f_{yy}(0, 0) - [f_{xy}(0, 0)]^2$

$\Rightarrow D = -4 - 0^2 = -4 < 0$

Hence, at (0, 0) f has a saddle point.

**Q23 Text Solution:**

Let us find the first-order partial derivative with respect to x and y to determine the critical points.

$f_x(x, y) = 2xy^2 - 10x - 8y$

and $f_y(x, y) = 2x^2y - 10y - 8x$ which exists everywhere.

Now, the second equation from first and equating to zero, we get

$2xy(y - x) + 2(y - x) = 0$

$\Rightarrow (y - x)(xy + 1) = 0$

$\Rightarrow$ either $x = y$ or $y = -1/x$

If $x = y$, substituting this in any of the partial derivatives, we get

$2x^3 - 18x = 0$

$\Rightarrow x = 0, 3, -3$

And if $y = -1/x$, substituting this in any of the partial derivatives, we get

$2/x - 10x + 8/x = 0 \Rightarrow 1 - x^2 = 0$

$\Rightarrow x = 1, -1$

Therefore, we get the critical points (0, 0), (3, 3), (−3, −3), (1, −1) and (−1, 1).

Now, $f_{xx}(x, y) = 2y^2 - 10$

$f_{yy}(x, y) = 2x^2 - 10$

$f_{xy}(x, y) = 4xy - 8$

At (0, 0), $D = f_{xx}(0, 0)\, f_{yy}(0, 0) - [f_{xy}(0, 0)]^2 = 100 - 64 = 36 > 0$ and $f_{xx}(0, 0) = -10 < 0$

Hence, at (0, 0) we have a local maximum

At (±3, ±3), $D = f_{xx}(\pm3, \pm3)\, f_{yy}(\pm3, \pm3) - [f_{xy}(\pm3, \pm3)]^2 = 64 - 784 = -720 < 0$

Hence, at (±3, ±3) we have saddle points.

At (±1, ±1), $D = f_{xx}(\pm1, \pm1)\, f_{yy}(\pm1, \pm1) - [f_{xy}(\pm1, \pm1)]^2$ = 64 − 144 = a negative quantity

Hence, at (±1, ±1) also we have saddle points.

**Q24 Text Solution:**

Since perpendicular distance of the point from curve is the minimum distance so maxima-minima concept can be applied here such that distance d is -

$$d = \sqrt{x^2 + y^2 + z^2}$$

$$z^2 = 3xy + 4$$

$$d = \sqrt{x^2 + y^2 + 3xy + 4}$$

$$f\left(x, y\right) = d^2 = x^2 + y^2 + 3xy + 4$$

$$\frac{\partial f}{\partial x} = 2x + 3y, r = \frac{\partial^2 f}{\partial x^2} = 2 \; and \; S = \frac{\partial^2 f}{\partial x \partial y} = 3$$

$$\frac{\partial f}{\partial y} = 2y + 3x$$

$$\frac{\partial^2 f}{\partial y^2} = 2$$

$$rt - s^2 = 4 - 3 > 0$$

$$r = 2 > 0$$

$$point \; of \; relative \; minima$$

$$f\left(x, y\right) = f\left(0, 0\right) = 4$$

$$d^2 = 4$$

$$d = 2$$

**Q25 Text Solution:**

Let P(x,y) be the position of jet and the soldier is placed at A(3,2) . Then, the distance between the soldier and jet is given by.

$$AP = \sqrt{(x-3)^2 + (y-2)^2}$$

$$= \sqrt{(x-3)^2 + x^4}$$

$$\left[\because \; y = x^2 + 2\right] \ldots\ldots\ldots (i)$$

Let $Z = AP^2$. Then $AP = \sqrt{Z}$ so $(AP)_{min} = \sqrt{Z_{min}} \ldots\ldots\ldots (2)$

$Z = (x - 3)^2 + x^4$

$$\frac{dz}{dx} = 0 \Rightarrow x = 1$$

$$\left(\frac{d^2z}{dx^2}\right)_{x=1} = -ve$$

so, Z is min at x = 1 & $Z_{min} = 5$

Hence, $AP_{min} = \sqrt{5}$

**Q26 Text Solution:**

Let x cum be the length of a side of the square which is cut-off from each corner of the plate.

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Then, sides of the box as shown in figure are 24 −2x, 24 −2x and x.

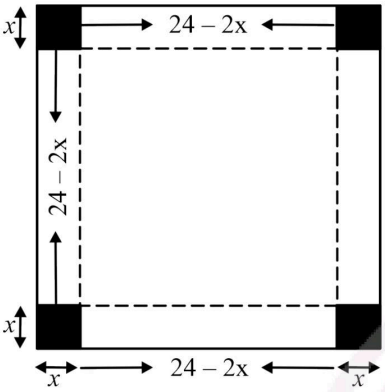
Let V be the volume of the box.

Then,

$V = (24 - 2x)^2 x = 4x^3 - 96x^2 + 576x$

$\Rightarrow \frac{dV}{dx} = 12x^2 - 192x + 576$ and $\frac{d^2V}{dx^2} = 24x - 192$

For maximum or minimum values of V, must have




$\frac{dV}{dx} = 0$

$\Rightarrow 12x^2 - 192x + 576 = 0$

$\Rightarrow x^2 - 16x + 48 = 0$

$\Rightarrow (x - 12)(x - 4) = 0 \Rightarrow x = 12, 4$

But, x = 12 is not possible.

Therefore, x = 4

Now, $\left(\frac{d^2V}{dx^2}\right) = 24 \times 4 - 192 < 0$.

Thus, V is maximum when x = 4.

Hence, the volume of the box is maximum when the side of the square is 4 cm.

The maximum volume is , $V = (24 - 8)^2 \times 4 = 1024$ $cm^3$

**Q27 Text Solution:**

Suppose x items are sold to maximize the profit P. Then, P=Revenue-Cost

$\Rightarrow P = x\left(5 - \frac{x}{100}\right) - \left(\frac{x}{5} + 500\right)$

$P = \frac{24}{5}x - \frac{x^2}{100} - 500$

$\frac{dP}{dx} = \frac{24}{5} - \frac{2x}{100}$

$\frac{d^2P}{dx^2} = \frac{-1}{50}$

The critical numbers of P are given by $\frac{dP}{dx} = 0$

$\frac{dP}{dx} = \frac{24}{5} - \frac{2x}{100} = 0$

$x = 240$

$\frac{d^2P}{dx^2} = \frac{-1}{50} < 0$

Hence, profit P is maximum when 240 items are sold.

**Q28 Text Solution:**

$\int x^2 \sin x \, dx$

$= x^2 \int \sin x \, dx - \int (2x \int \sin x \, dx) dx$

$= -x^2 \cos + 2 [x \int \cos x \, dx -$

$\int (1 \int \cos x \, dx) \, dx]$

$= -x^2 \cos x + 2x \sin x + 2 \cos x + c$

**Q29 Text Solution:**

$x = t^2 \Rightarrow dx = 2t \, dt$

$I = \int \frac{dx}{2\sqrt{x}(x+1)} = \int \frac{2t \, dt}{2t(t^2+1)}$

$\int \frac{dt}{1+t^2} = \tan^{-1} t + c$

$= \tan^{-1} (\sqrt{x}) + c$

**Q30 Text Solution:**

Put

$\frac{1}{(x-1)(x-2)(x-3)} = \frac{A}{(x-1)} + \frac{B}{(x-2)} + \frac{C}{(x-3)}$

$\Rightarrow$ I = A (x − 2)(x − 3) +B (x − 1) (x − 3) + C(x − 1) (x − 2)

Put $x = 1$, we get, $A = \frac{1}{2}$

$x = 2$, we get, $B = -1$

$x = 3$, we get, $C = \frac{1}{2}$

So integral, $= \frac{1}{2} \int \frac{dx}{x-1} - \int \frac{dx}{x-2} + \frac{1}{2} \int \frac{dx}{x-3}$

$= \ln \left( \frac{\sqrt{x^2-4x+3}}{|x-2|} \right) + c$

**Q31 Text Solution:**

$\int_2^7 \frac{\sqrt{x} dx}{\sqrt{x} + \sqrt{9-x}}$

$I = \int_2^7 \frac{\sqrt{9-x}}{\sqrt{9-x} + \sqrt{9-(9-x)}} dx$

$I = \int_2^7 \frac{\sqrt{9-x}}{\sqrt{9-x} + \sqrt{x}} dx$

Adding (i) and (ii), we get

$2I = \int_2^7 \left( \frac{\sqrt{x}}{\sqrt{x} + \sqrt{9-x}} + \frac{\sqrt{9-x}}{\sqrt{x} + \sqrt{9-x}} \right) dx$

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$= \int_2^7 dx = x\Big|_2^7 = 5$

$I = \frac{5}{2}$

**Q32 Text Solution:**

$I = \int_{-\sqrt{3}}^{\sqrt{3}} \frac{dx}{(1+e^x)(1+x^2)}$

Here, $f(x) = \frac{1}{(1+e^x)(1+x^2)}$

$\Rightarrow f(-x) = \frac{1}{(1+e^{-x})\left(1+(-x)^2\right)} = \frac{e^x}{(1+e^x)(1+x^2)}$

So, $I = \int_0^{\sqrt{3}} \frac{dx}{1+x^2} = \tan^{-1} x\Big|_0^{\sqrt{3}} = \frac{\pi}{3}$

**Q33 Text Solution:**

$I = \int_{-a}^{a} \sqrt{\frac{a-x}{a+x}}dx = \int_{-a}^{a} \frac{a-x}{\sqrt{a^2-x^2}}dx$

$= a\int_{-a}^{a} \frac{dx}{\sqrt{a^2-x^2}} - \int_{-a}^{a} \frac{x\,dx}{\sqrt{a^2-x^2}}$

$= a \cdot 2\int_0^a \frac{dx}{\sqrt{a^2-x^2}}$

$- 0 \left(\because \frac{x}{\sqrt{a^2-x^2}} \text{is an odd function}\right)$

$= 2a\left[\sin^{-1}\frac{x}{a}\right]_0^a$

$\Rightarrow 2a\left[\sin^{-1}(1) - \sin^{-1}(0) = a\right]$

$= a\left[\frac{\pi}{2} - 0\right] = \pi a$

**Q34 Text Solution:**

$I = \int_0^{\pi} \frac{x dx}{1+\cos^2 x}$

$I = \int_0^{\pi} \frac{(\pi-x)dx}{1+\cos^2(\pi-x)} = \int_0^{\pi} \frac{(\pi-x)dx}{1+\cos^2 x}$

Addition both, we get

$2I = \int_0^{\pi} \frac{\pi\,dx}{1+\cos^2 x} \Rightarrow I = \frac{\pi}{2}\int_0^{\pi} \frac{dx}{1+\cos^2 x}$

$= \frac{\pi}{2}\left[\int_0^{\pi} \frac{dx}{1+\cos^2 x} + \int_0^{\pi/2} \frac{dx}{1+\cos^2(\pi-x)}\right]$

$= \pi\int_0^{\pi/2} \frac{\sec^2 x\,dx}{2+\tan^2 x}$ Put $\tan x = t$

$I = \pi\int_0^{\infty} \frac{dt}{t^2+2}$

$= \frac{\pi}{\sqrt{2}}\tan^{-1}\left(\frac{t}{\sqrt{2}}\right)\Big|_0^{\infty} = \frac{\pi^2}{2\sqrt{2}}$

**Q35 Text Solution:**

$\int_0^{\pi} \frac{dx}{1+2\sin^2 x} = 2\int_0^{\pi/2} \frac{dx}{1+2\sin^2 x}$

$\Big($

$\int_0^{2a} f(x)dx = 2\int_0^a f(x)dx,\ if\ f(2a-x)$

$= f(x)\Big)$

$= 2\int_0^{\pi/2} \frac{\sec^2 x dx}{\sec^2 x + 2\tan^2 x} = 2\int_0^{\pi/2} \frac{\sec^2 x dx}{1+3\tan^2 x}$

(Not that in the beginning can not divided numerator and denominator by $\cos^2 x$ , as $\cos x = 0$ at $x = \pi/2$)

$= 2\int_0^{\infty} \frac{dt}{1+3t^2}, \qquad (\tan x = t)$

$= 2\frac{1}{\sqrt{3}}\left[\tan^{-1} t\sqrt{3}\right]_0^{\infty} = \frac{2}{\sqrt{3}} \times \frac{\pi}{2} = \frac{\pi}{\sqrt{3}}$

**Q36 Text Solution:**

$5f(x) + 4f\left(\frac{1}{x}\right) = \frac{1}{x} + 3$----------

(1)

Replace $x \to \frac{1}{x}$

$5f\left(\frac{1}{x}\right) + 4f(x) = x + 3$----------

(2)

Multiplyb equation 1 with 5 and multiply equation 2 with 4 and then subtract bith the equation we get-

$f(x) = \frac{1}{9}\left(\frac{5}{x} - 4x + 3\right)$

$\therefore I = 18\int_1^2 \frac{1}{9}\left(\frac{5}{x} - 4x + 3\right)dx$

$I' = 18\int \frac{1}{9}\left(\frac{5}{x} - 4x + 3\right)dx$

$2\left(5\log x - 2x^2 + 3x\right)$

$10\log x - 4x^2 + 6x$

*Putting the limits*

$10\log 2 - 10\log 1 - 4\left(4-1\right) + 6\left(2-1\right)$

$10\log 2 - 12 + 6$

$10\log_e 2 - 6$

**Q37 Text Solution:**

$F'(x) = 2x \cdot \sqrt{\sin x^2} - 1 \cdot \sqrt{\sin x}$

**Q38 Text Solution:**

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$$\lim_{x\to\infty}\frac{\left(\int_0^x e^{t^2}\,dt\right)^2}{\int_0^x e^{2t^2}\,dt}\because\left(\frac{\infty}{\infty}form\right)$$

Applying L's Hospital rule:

$$\lim_{x\to\infty}\frac{2\cdot\int_0^x e^{t^2}\,dt\cdot e^{x^2}}{1\cdot e^{2x^2}}$$

$$\lim_{x\to\infty}\frac{2\cdot\int_0^x e^{t^2}\,dt}{e^{x^2}}=\lim_{x\to\infty}\frac{2\cdot e^{x^2}}{2x\cdot e^{x^2}}$$

$$\lim_{x\to\infty}\frac{1}{x}=0$$

**Q39 Text Solution:**

Arc length $=\int_0^4\sqrt{1+\left(\frac{dy}{dx}\right)^2}dx$

$=\int_0^4\sqrt{1+\frac{9x}{4}}dx$

$=\frac{8}{27}\left(1+\frac{9}{4}x\right)^{3/2}\Big|_0^4=\frac{8}{27}\left(10^{3/2}-1\right)$

**Q40 Text Solution:**

$L=\int_1^2\sqrt{1+\frac{1}{4x^2}}dx$

$\int_1^2\sqrt{1+\left(\frac{dy}{dx}\right)^2}dx=\int_1^2\sqrt{1+\frac{1}{4x^2}}dx$

So, $\frac{dy}{dx}=\frac{1}{2x}$

$\int dy=\frac{1}{2}\int\frac{dx}{x}+c$

$y=\ln\sqrt{x}+c$

y = f(x) passes through (1, 1)

$y=\ln\sqrt{x}+1\Rightarrow c=1$

**Q41 Text Solution:**

Volume of solid of revolution $V=\int_0^b\pi y^2dx$

$=\frac{\pi a^2}{4}\int_0^b\left(e^{2x/a}+e^{-2x/a}+2\right)dx$

$=\frac{\pi a^2}{4}\left[\frac{a}{2}e^{2x/a}-\frac{a}{2}e^{-2x/a}+2x\right]_0^b$

$=\frac{\pi a^3}{8}\left(e^{2b/a}-e^{-2b/a}\right)+\frac{\pi a^2 b}{2}$

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